Chain-of-Thought (CoT), Zero-Shot CoT, and Reasoning Models — Detailed Notes

1. Introduction

Chain-of-Thought (CoT) prompting is a technique used with Large Language Models (LLMs) to improve their ability to solve problems that require reasoning. Instead of producing a direct answer, the model is guided to generate intermediate reasoning steps before arriving at the final solution. This approach became important because early language models were good at generating fluent text but weak at logical and multi-step reasoning tasks.

2. Limitations of Early Language Models

Early LLMs (around 2022) performed well on tasks involving memorization, pattern recognition, and language generation. However, they struggled with tasks that required structured thinking, such as mathematics, logical puzzles, and multi-step reasoning problems.

The main issue was that these models tended to produce answers immediately without working through the problem. Even increasing model size did not significantly improve reasoning ability. As a result, performance on reasoning benchmarks remained limited.

3. Concept of Chain-of-Thought Prompting

Chain-of-Thought prompting addresses this limitation by encouraging the model to break a problem into smaller steps and reason through them sequentially.

Instead of:

Problem → Answer

CoT uses:

Problem → Reasoning Steps → Final Answer

This mirrors how humans solve complex problems by working through intermediate steps.

4. How CoT Prompting Works

CoT prompting is implemented by instructing the model to explain its reasoning before giving the answer. This can be done by:

• Asking the model to think step by step
• Requesting an explanation of the reasoning process
• Providing examples that show step-by-step solutions

When guided in this way, the model generates a sequence of logical steps that lead to a more accurate solution.

5. Few-Shot Chain-of-Thought Prompting

Few-shot CoT involves giving the model several examples of problems along with their detailed solutions. The model learns the reasoning pattern from these examples and applies it to new problems.

This method is effective but requires carefully prepared examples and longer prompts.

6. Zero-Shot Chain-of-Thought Prompting

Zero-Shot CoT does not require examples. Instead, a simple instruction such as “Let’s think step by step” is added to the prompt. This phrase alone can significantly improve reasoning performance.

The effectiveness of Zero-Shot CoT suggests that language models already contain latent reasoning abilities that can be activated through appropriate instructions.

7. Intermediate Reasoning as a Scratchpad

CoT provides the model with a textual workspace where it can perform calculations and logical operations before producing the final answer. This intermediate reasoning reduces mistakes and improves consistency.

8. Transition to Modern Reasoning Models

Recent models are increasingly trained to perform reasoning internally. These models can generate answers that reflect a reasoning process even without explicit CoT prompts.

Advances contributing to this shift include:

• Training on reasoning-focused datasets
• Reinforcement learning techniques

• Self-consistency methods
• Planning and tool-use training

As a result, modern models often produce better reasoning outputs by default.

9. Current Relevance of CoT

Chain-of-Thought prompting is still useful in several situations:

• When using smaller or less advanced models
• When detailed explanations are required.
• For complex problems where explicit reasoning helps verification

However, for highly advanced reasoning models, explicit CoT prompting may not always be necessary.